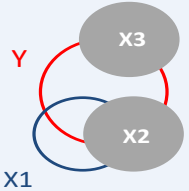


Topics Guide: The Taxonomy of Control Variables

Which controls fix which problems, what goes wrong when we leave them out, and what new problems appear when a control is too closely tied to the policy variable. If you can explain each bullet in your own words and work the bias calculation, you understand the unit.

Key formulas worth remembering	
Full model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$ Naïve model: $Y = b_0 + b_1 X_1 + e$ (X_2 omitted) Auxiliary model: $X_2 = \alpha_0 + \alpha_1 X_1 + \varepsilon_2$ (omitted variable on the left) SE of the slope $\approx \text{residual} / (n \cdot \text{var}(X_1))$	Naïve slope = true slope + bias: $b_1 = \beta_1 + \alpha_1 \beta_2$ Omitted variable bias $= \alpha_1 \times \beta_2 = (X_1 \rightarrow X_2) \times (X_2 \rightarrow Y)$ Direction of bias = $\text{sign}(\alpha_1) \times \text{sign}(\beta_2)$ Variance inflation: $VIF = 1 / (1 - r^2_{12})$; SE of b_1 grows by $\sqrt{VIF}$



B1 = program impact

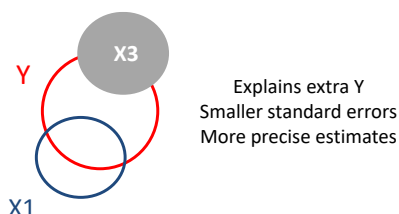
Full Model: $Y = B_0 + B_1 \cdot X_1 + B_2 \cdot X_2 + B_3 \cdot X_3 + e$

policy variable (points to X_1)

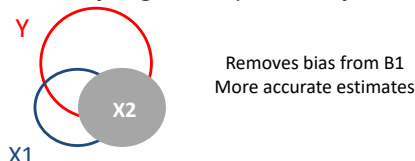
controls (points to X_2 and X_3)

01 · Two Goals, Two Kinds of Control

- **A good estimate of program impact is accurate and precise.** Accurate = unbiased (no omitted variable bias). Precise = efficient (small standard errors). Separate goals with separate remedies.
- **Every control predicts Y — that is what makes it a control.** What sorts it into a type is its correlation with the *policy variable* X_1 , not with Y .
- **Type A: uncorrelated with X_1 (X_3).** A pure predictor of Y . It explains extra variance in Y and shrinks the residual: **slope unchanged, SE smaller**. It buys **precision (power)** almost for free.



- **Type B: correlated with X_1 (X_2).** A competing hypothesis. It takes a share of the $X_1 \rightarrow Y$ overlap: **slope changes, bias removed, SE usually larger**. It buys **accuracy**.



- **Precision is a technical fix; accuracy is a claim about the world.** You can hunt for Type A controls and watch the SE shrink. Type B controls come from theory — no diagnostic tells you which competing hypothesis is missing.

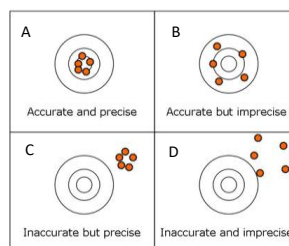
Which control fixes which problem

Control	Corr. with X_1 ?	Fixes	Slope b_1	SE of b_1
Type A: predicts Y	no	power (precision)	unchanged	smaller
Type B: predicts Y	yes	bias (accuracy)	moves toward β_1	usually larger
Related to X_1 but not Y	yes	nothing	unchanged	larger (pure cost)
Related to neither	no	nothing	unchanged	$\approx$ unchanged

02 · Omitting Controls: The Accuracy/Precision 2 × 2

	X_3 included	X_3 omitted
X_2 included	A · unbiased & precise	B · unbiased & imprecise
X_2 omitted	C · biased & precise	D · biased & imprecise

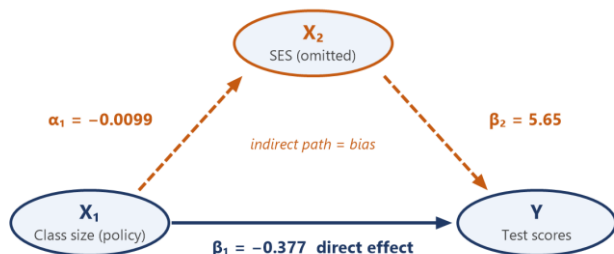
- **A is the goal; B is honest.** Omitting a Type A control leaves the slope right but widens the CI — a risk of a false negative.
- **C is the dangerous cell: biased with false confidence.** Tight CIs around the wrong number. Small SEs, strong significance, and high R^2 cannot see bias. The naïve model D is also biased, but at least it advertises its uncertainty.



- **Adding noise-reducing controls to a mis-specified model makes it look better without making it better.** Power (precision) should be a secondary consideration to bias (accuracy).

03 · Omitted Variable Bias: Direction and Size

- **Greek = truth, Latin = our guess.** The full model gives β_1 ; the naïve model gives b_1 . Bias = $b_1 - \beta_1$, which requires the truth — so we use the path diagram instead.
- **Bias is the indirect effect.** b_1 = direct + indirect effect; β_1 = direct effect. The naïve slope absorbs the path $X_1 \rightarrow X_2 \rightarrow Y$ because it cannot tell the two routes apart.



Path diagram from the class-size example (Y = test scores, X_1 = class size, X_2 = SES).

- **Bias = $\alpha_1 \times \beta_2$, so both links must be open:** $\alpha_1 \neq 0$ (omitted variable moves with X_1) and $\beta_2 \neq 0$ (it affects Y). If either is zero, there is no bias. **Size** depends on the strength of both links; **direction** is the product of their signs.

Worked example (class size $\rightarrow$ test scores, SES omitted)

Step	Calculation
Naïve model	$b_1 = -0.433$
Full model	$\beta_1 = -0.377, \beta_2 = 5.65$ (SES $\rightarrow$ scores)
Auxiliary model	$\alpha_1 = -0.0099$ (class size $\rightarrow$ SES)
Bias = $\alpha_1\beta_2$	$(-0.0099)(5.65) = -0.056$
Check: $b_1 - \beta_1$	$-0.433 - (-0.377) = -0.056 \checkmark$

- **Reading the example:** bigger classes have lower-SES students ($\alpha_1 < 0$) and SES raises scores ($\beta_2 > 0$), so the bias is negative. The naïve model makes class size look about 15% more harmful than it is ($0.056 / 0.377$). Omitting SES and teacher quality together overstates the impact by 51%.

Signing the bias from the path diagram

$\alpha_1 (X_1 \rightarrow X_2)$	$\beta_2 (X_2 \rightarrow Y)$	Bias	Naïve slope b_1 is
+	+	+	too large
-	-	+	too large
+	-	-	too small
-	+	-	too small

- **You can sign the bias from theory without measuring X_2 .** If the naïve slope is too large, a program can look significant when it is not; if the bias works against you, the true effect is

bigger than reported. With a negative effect, “too small” means *more negative* — larger in magnitude.

- **Bias does not shrink with sample size.** It is a missing *column* problem, not a missing *row* problem.

04 · When Correlations Are Too High: Multicollinearity

- **The price of fixing bias.** When a Type B control is highly correlated with X_1 , the circles overlap almost completely. The shared region is discarded, so b_1 is estimated from the thin slice of X_1 that varies independently of X_2 .
- **Standard errors inflate.** SE of b_1 grows by $\sqrt{\text{VIF}}$: $r = 0.5 \rightarrow \times 1.15$; $r = 0.9 \rightarrow \times 2.3$; $r = 0.95 \rightarrow \times 3.2$; $r = 0.99 \rightarrow \times 7.1$. Real effects can lose significance (lost power).
- **Slopes become unstable.** With little independent variation, dropping a few cases, adding a variable, or drawing a new sample can swing b_1 and b_2 widely, even flipping signs. The model cannot decide which variable deserves the credit.
- **Still unbiased.** Multicollinearity is a precision problem, not an accuracy problem: right on average, noisy in any one sample.
- **Warning signs:** a significant model (high R^2 , joint F-test) with individually insignificant coefficients; large changes in b_1 when X_2 is added; $r(X_1, X_2)$ above ~ 0.8 or VIF above ~ 5 –10.
- **Do not drop the confounder just to shrink the SE** — that trades imprecision for bias (cell C). Instead: more data, more variation in X_1 at the design stage, combine near-duplicate controls into an index, or ask whether the control measures the same thing as the policy. At $r = 1$ the model cannot be estimated at all.

Cheat sheet: what each specification choice does

Situation	Slope b_1	SE of b_1
Add Type A control (uncorrelated with X_1)	unchanged	smaller
Add Type B control (correlated with X_1)	changes (bias removed)	usually larger
Omit Type A control	unbiased	larger
Omit Type B control	biased by $\alpha_1\beta_2$	often smaller (false confidence)
Control very highly correlated with X_1	unbiased but unstable	much larger ($\times \sqrt{\text{VIF}}$)
Control related to X_1 but not Y	unchanged	larger (pure cost)
Randomized X_1	unbiased ($\alpha_1 = 0$)	unchanged

Two questions for every control: Is it correlated with the policy variable (bias)? Does it predict the outcome (precision)?